

UNIVERSITY OF AMSTERDAM

MASTERS THESIS

The Origins of Order: Strategic Reasoning, Information, and Enforcement in Multi-Agent LLM Societies

Author:

Koen VERLAAN

Supervisor:

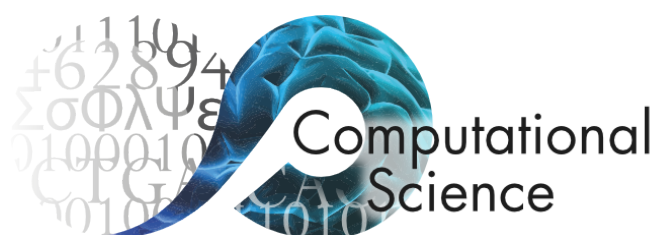
Dr. Debraj Roy

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Chapter 3

Methods

3.1 Coordination Game Design

Our game is designed to produce rich emergent dynamics by utilizing minimally sufficient components, following the *Generative Sufficiency* precedent set by Epstein (2006). The step away from classical games researched thoroughly in game theory, such as the Prisoners Dilemma or Stag Hunt, is motivated by the absence of emergent behaviour possible using large-language model driven agent architectures. The six actions prove to be rich enough for dynamics such as coalition-forming, arms races, bandwagoning, mobbing, hierarchies and signalling, where classical games simply wouldn't suffice without modifications.

Our coordination game is defined as a tuple $G = \langle \mathcal{N}, \mathcal{S}, \mathcal{A}, O, T, u \rangle$:

- $\mathcal{N} = \{1, \dots, n\}$: a set of $n = 30$ agents.
- $\mathcal{S} = (\mathbf{R}(t), \mathbf{B}(t), G(t))$: the state at round t , consisting of resource vectors $\mathbf{R}(t) \in \mathbb{R}_+^n$, arm bonus vectors $\mathbf{B}(t) \in \mathbb{R}_+^n$, and a social network $G(t) = (V, E(t))$ where $V = \mathcal{N}$ and the edge set $E(t)$ evolves through rewiring (Section 3.1.1).
- $\mathcal{A} = \{\text{do_nothing}, \text{invest_other}(j), \text{arm_self}, \text{arm_other}(j), \text{attack}(j)\}$: the action set, where $j \in \mathcal{N}_i(t)$ denotes a network neighbour of agent i . Agents may only target direct connections.
- $O_i(t) \subseteq \mathcal{S}$: the observation function. Agent i observes its own resources and arm bonus, but *not* the resources of neighbours. Agent i observes the recent actions and combat outcomes involving its network neighbourhood $\mathcal{N}_i(t)$, accumulated through persistent local memory (Section 3.2.3).

- $T : \mathcal{S} \times \mathcal{A}^n \rightarrow \mathcal{S}$: the transition function. All agents act simultaneously; actions are resolved in a single step. All costs and returns are percentages of the acting agent’s current resources.
- $u_i(t) = R_i(t+1) - R_i(t)$: the payoff for agent i , defined as the change in resources. Agents are instructed to maximise their absolute resource count.

The game is played by 30 homogeneous LLM-driven agents. Section 3.2 explores the exact architecture and rationale behind choosing Qwen 3.5-72B as model. The amount of agents in our simulations was bound by two facts: the most-cited LLM-driven agent-based model used 25 agents and proved enough for rich emergent social structures. The number also falls in the range to detect communities in small networks. Additionally, a single H100 is able to process 32 concurrent sequences in vLLM. Our choice of 30 sits rounded in the middle.

The state S is defined by two vectors and a network. Resources $R_i(t)$ is a vector of non-negative reals and is initialized for all agents at $R_i = 100$. Arm bonuses $B_i(t)$ are also tracked through a vector and represent an agent’s strenght during combat when summed with their resources. Every round, all arm bonuses decay with δ .

The network For a detailed explanation of the network topology, see 3.1.1.

All costs and returns are a percentage of the actor’s resources $R_i(t)$. This design choice rests on assuring scale invariance, meaning both rich and poor agents face similar dilemmas disregarding their wealth and aligns with canonical ABM practice.

There are two **cooperative** actions, **invest_other** and **invest_self**, which grow the total economy. However, investing in another agent costs the actor $c_{inv}\%$ of their resources, making the action only valuable for their own wealth if they expect reciprocity in the following rounds. Investing in oneself removes that uncertainty, but the net percentage gain (g_{self}) is smaller and the actor is limited by their own wealth. Both actions simultaneously exhibit a signalling function: investing in others can be seen as goodwill, trust and an invitation for cooperation, while investing in oneself can be seen as asocial, though practical. Additionally, the cost of investing in another agent (c_{inv}) is smaller than the return for the target (r_{inv}), creating a social dilemma: individually investing is costly, but collectively the action is net-positive.

Three **military** actions introduce additional dynamics: **arm_self** creates extra combat power (B_i) for a limited time and can be used both offensively and defensively. Seeing a neighbor arm themselves poses a threat either way, as their intentions are not known to the agent. Actors can also decide to **arm_other**, increasing the combat power of a neighbor. Arming other agents can signal extreme trust and enables stronger agents to

substantially strengthen a weaker neighbor for a limited time. The combat bonuses are subject to a decay (δ) to prevent permanent escalation. Keeping high combat bonuses requires repeated arming.

The driving force in the game is the **attack** action. Both the actor and target pay conflict cost (c_{war}) after which their chances of winning the conflict is decided by the ratio of their resources R_i + arm bonus B_i . Conflicts are probalistically resolved, meaning the stronger agent is more probable to win than the weaker one. The winner takes $\alpha\%$ of the losers resources, irrespective to who was the initiator of the attack. When multiple agents attack a single target in same round, either through cooperation or chance, their combat strength combines into a single coalition attack. If the defender loses the conflict, $\alpha\%$ will be split proportionally by strenght contribution. Defenders always stand alone in our game. Arming other becomes the effective mechanism for strengthening coalition partners defensively.

Because all actions happen simulatenously, an agent involved in multiple conflicts wil use their snapshotted strength in combat. An agent can lose at most

Lastly, a single **passive** action allows agents to do nothing in any given round. Although the action itself costs nothing and gains nothing in terms of resources, the loss of opportunity is to be considered.

Agents are able to fully observe their own resources R_i and arm bonus B_i . On default, the resources of their neighbours are hidden, creating type uncertainty. Actions and combat outcome involving direct neighbours are visible and accumulate in memory (see [3.2.3](#))

3.1.1 Interaction Topology

As far as I understand, graphs are initialized using Erdos-Reyni. Is it not unfair that agents potentially only drop 1 connection and gain one? That means agents can never create more connections than theyre initialized with? Should the reconnecting also be smart? Or is connecting to a random non-neighbor automatically beneficial to lesser connected agents?

TABLE 3.1: Game parameters. All action costs and returns are percentages of the acting agent’s current resources $R_i(t)$. Default values are used unless explicitly varied.

Action	Effect	Parameter	Default
<code>invest_other(j)</code>	Agent pays $c_{\text{inv}}\%$ of R_i ; target j receives $r_{\text{inv}}\%$ of R_i . Grows the total economy.	$c_{\text{inv}} / r_{\text{inv}}$	10% / 15%
<code>invest_self</code>	Agent gains $g_{\text{self}}\%$ of R_i (no cost).	g_{self}	2%
<code>arm_self</code>	Agent pays $c_{\text{arm}}\%$ of R_i (removed from economy); combat bonus B_i increases by $c_{\text{arm}} \cdot m$. Bonus decays by δ each round.	$c_{\text{arm}} / m / \delta$	10% / 2.0 / 0.5
<code>arm_other(j)</code>	Agent pays $c_{\text{arm},j}\%$ of R_i (removed from economy); target j ’s combat bonus B_j increases by $c_{\text{arm},j} \cdot m$.	$c_{\text{arm},j}$	10%
<code>attack(j)</code>	Both pay $c_{\text{war}}\%$ of own resources. Combat strength $s_i = R_i + B_i$. If multiple agents attack the same target, their strengths combine. Win probability $p = \sum s_{\text{atk}} / (\sum s_{\text{atk}} + s_{\text{def}})$. Winner takes $\alpha\%$ of loser’s resources; coalition spoils split proportionally by strength.	c_{war} / α	5% / 40%
<code>do_nothing</code>	No cost, no effect.	—	—
Simulation	Description	Parameter	Default
	Number of agents	n	30
	Initial resources per agent	$R_i(0)$	100
	Hard cap (with early stopping)	<code>max_rounds</code>	50
	LLM sampling temperature	T	0.7

3.2 Agent Architecture and Model Selection

3.2.1 Model Selection

3.2.2 Prompt Structure

3.2.3 Agent Memory

3.3 Independent Variables

3.3.1 Reasoning Depth

3.3.2 Information Structure

3.3.3 Communication Scope

3.4 Experimental Design

3.4.1 Convergence Detection

3.4.2 Parameter Characterization (OAT Screening)

3.4.3 Reasoning Depth Experiments

3.4.4 Information Structure Experiments

3.4.5 Communication Scope Experiments

3.4.6 Robustness Checks

3.5 Metrics

3.5.1 Primary Metrics

3.5.2 Secondary Metrics

3.5.3 Statistical Analysis

3.6 Computational Infrastructure